

Definition and first terms

$5\,A(x) = 4 + 4\,x + A(x)^3$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(2)*T(x)$; $T=x+3*T^2+2*T^3$.
 $a(0),\ldots,a(7) = 1, 2, 6, 40, 330, 3048, 30156, 312528$.
 Kernel used throughout: $\rho(u) = u\,(1-3\,u^1-2\,u^2)$, so $x = \rho(T(x))$.

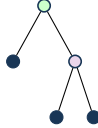
Typogeometry



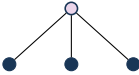
1x2



{1,1}x6



{1,{1,1}}x18



{1,1,1}x4

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 3, Δ_3 : 2.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1,\ldots,k_2) \in \mathbb{Z}_{\geq 0}^2 : \sum_{j=1}^2 j k_j = n-1\}, \quad K = \sum_{j=1}^2 k_j,$$

$$a(n) = \frac{2}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1,\ldots,k_2} 3^{k_1} 2^{k_2}.$$

$$a(n) = \frac{2}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^2 P_r(n) a(n+r) = 0, \; n \geq 1, \; \text{with}$$

$$\begin{aligned}
 P_0(n) &= -108\,n^2 + 12 \\
 P_1(n) &= -216\,n^2 - 324\,n - 108 \\
 P_2(n) &= 17\,n^2 + 51\,n + 34
 \end{aligned}$$

Differential equation

$$(12\,x^2)\,A(x) + (-108\,x^3 - 108\,x^2)\,A'(x) + (-108\,x^4 - 216\,x^3 + 17\,x^2)\,A^{(2)}(x) = 0.$$

Matrices, telescoper certificate, and checks

$$U = \begin{pmatrix} \frac{198}{17} & \frac{30}{17} & \frac{3}{17} \\ \frac{180}{17} & \frac{18}{17} & \frac{12}{17} \\ 0 & 0 & 0 \end{pmatrix}, \quad
 V = \begin{pmatrix} -1 & 0 & 0 \\ \frac{96}{17} & \frac{13}{17} & \frac{3}{17} \\ \frac{60}{17} & \frac{6}{17} & \frac{4}{17} \end{pmatrix}, \quad
 J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Certificate.

$$R(n,u) = \frac{-72\,n\,u^4 - 144\,n\,u^3 + 42\,n\,u^2 + 114\,n\,u - 17\,n - 24\,u^4 - 48\,u^3 - 6\,u^2 + 6\,u}{\rho(u)^1}.$$

Telescoping identity: $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$.

Matrix dimensions: 6×6 ; remainder 2×3 , rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.